

Identification of acquired coagulation disorders and effects of target-controlled coagulation factor substitution on the incidence and severity of spontaneous intracranial bleeding during veno-venous ECMO therapy

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Abstract

Introduction: Intracranial haemorrhage is a redoubtable complication during extracorporeal membrane oxygenation (ECMO) therapy. The underlying mechanisms of haemorrhagic diathesis are still not completely understood. This study was performed to evaluate a coagulation protocol for the regular analysis of acquired coagulation disorders and the systematic substitution of coagulation factors to reach predefined target values. We hypothesised that using this strategy would lead to the identification of acquired bleeding disorders which cannot be monitored with standard coagulation tests and that substitution of the respective factors in a target-controlled approach could have an impact on the incidence and severity of intracranial haemorrhage.

Methods: A protocol for the analysis of acquired coagulation disorders and the subsequent administration of associated factor concentrates was introduced. Previously, coagulation management was mainly based on clinical bleeding signs as the trigger for the administration of blood products. In this investigation, nineteen consecutive patients before (control group) and twenty consecutive patients after the implementation of the protocol (intervention group) have been included in the study.

Results: Eighty-eight percent of the patients developed factor XIII deficiency, 79% acquired von Willebrand syndrome, 40% fibrinogen deficiency and 54% of the patients showed a decline in platelet count >20% within the first 24 hours of ECMO therapy. In 6 out of 19 (31%) patients in the control group and in 2 patients out of 20 (10%) in the intervention group, intracranial haemorrhage was detected. Whilst 5 of 6 patients in the control group died because of fatal bleeding, both of the patients in the intervention group recovered with a favourable neurologic outcome.

Conclusions: Veno-venous ECMO therapy leads to thrombocytopenia, factor XIII and fibrinogen deficiency as well as acquired von Willebrand syndrome. The implementation of a coagulation protocol including a standardized determination and target-controlled substitution of coagulation factors may have a beneficial impact on the incidence and severity of intracranial haemorrhage.

Keywords

ARDS, ECMO, intracranial haemorrhage, bleeding, coagulopathy, factor XIII, von Willebrand syndrome, fibrinogen

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Introduction

Veno-venous extracorporeal membrane oxygenation (ECMO) therapy currently enjoys a renaissance in the rescue therapy of severe respiratory failure. Some authors already proclaim ECMO as an alternative to invasive mechanical ventilation.¹ ECMO may improve the outcome of severe acute respiratory distress syndrome (ARDS), but the definite indications, especially the question of when to start extracorporeal therapy, remains uncertain.² In particular, the increased risk of fatal bleeding complications, including intracranial haemorrhage, currently impede ECMO becoming routine in the treatment of severe ARDS.³ More than 30% of all ECMO patients suffer from relevant bleeding.⁴ While bleeding from cannula insertion sites, epistaxis and gastrointestinal bleeding are controllable complications in most cases, intracranial haemorrhage is reported with an incidence ranging from 5%¹ up to 19%,⁵ with predominantly fatal outcome. Data from the registry of the Extracorporeal Life Support Organisation (ELSO) confirm that bleeding generally is still the main problem during ECMO therapy; specifically, an incidence of 3.9% for intracranial haemorrhage with a survival rate of only 20% is reported.⁶ In a series in Australia and New Zealand in 2009, haemorrhage accounted for 71% of deaths, with intracranial haemorrhage being the cause of death in 60% of the patients who died.⁷

Clinical symptoms of intracranial bleeding may be absent or masked by the use of sedatives. Even misinterpretation of the symptoms as delirium is possible. The only way to prove intracranial bleeding during ECMO therapy is cranial computed tomography (CCT) which requires patient transport and additional harm. Therefore, it is likely that the estimated number of unidentified cases is much higher than the 3.9% reported in the retrospective ELSO Registry.

The mechanisms of ECMO leading to haemorrhagic diathesis are under active investigation. The development of acquired von Willebrand syndrome (AVWS) and the loss of high molecular weight von Willebrand multimers during the course of ECMO therapy has recently been demonstrated.⁸ In addition, pump-induced platelet injury leading to thrombocytopenia and impaired platelet function are also proposed to be reasonable factors.⁹ The combination of platelet dysfunction, acquired plasmatic coagulation disorders and anticoagulation with heparin is blamed for spontaneous intracranial haemorrhage in patients with ventricular assist devices¹⁰ and veno-arterial ECMO¹¹ as well.

In daily clinical practice, blood products are regularly substituted when signs of bleeding, such as diffuse nasal or oral mucosal haemorrhage or at puncture sites, occur. Fresh frozen plasma is widely used to control bleeding disorders, requiring up to 14 plasma units per ECMO day.^{9,12}

This study evaluates a protocol for the scheduled determination of acquired bleeding disorders and the substitution of blood products as well as coagulation factor concentrates to predefined target values, even in the absence of clinical bleeding signs. This target-controlled regimen is compared to a coagulation management performed previously, which was based mainly on clinical bleeding signs as a trigger for the administration of blood products. Special regard was paid to the incidence and severity of intracranial bleeding and the amount of applied red cell and platelet concentrates as well as fresh frozen plasma and other coagulation factors in patients undergoing veno-venous ECMO therapy.

Methods

Patients and data collection

The local ethics committee (Ethics Committee of Albert-Ludwigs-University Freiburg) approved the study and confirmed that no patient consent was required because of the retrospective, descriptive and anonymous design (EK162/13, April 19th 2013). All patients included in the study were treated in the Intensive Care Unit (ICU) at Freiburg University Medical Centre, Department of Anaesthesiology and Critical Care Medicine from 2009 till 2012. A protocol for the analysis of acquired coagulation disorders in patients undergoing ECMO was introduced in the department in July 2011 (Table 1). Nineteen consecutive patients before (control group) and twenty consecutive patients after (intervention group) the implementation of the abovementioned protocol have been included. Patients receiving veno-arterial extracorporeal support (extracorporeal life support; ECLS) were excluded. ECMO was implanted as a rescue therapy because of severe hypoxic failure despite conventional treatment when the mortality risk was estimated to be 80% or higher. Electronic ICU charts, ECMO implantation reports and laboratory results were used to collect all the required information.

Extracorporeal System

The patients were connected to ECMO by either two single-lumen HLS[®] catheters (Maquet Holding GmbH & Co. KG, Rastatt, Germany) or a transjugular two-stage Avalon Elite[®] catheter (Avalon Laboratories, Rancho Dominguez, California, USA). In four patients, a NovaPort Twin[®] cannula (Novalung GmbH, Heilbronn, Germany) was inserted. A Revolution[®] centrifugal pump (Sorin Group, Mirandola, Italy) and either an EOS[®] (Sorin Group; in 36 patients) or a Hilite[®] Oxygenator (Medos Medizintechnik AG, Stolberg, Germany, in three patients) was used for extracorporeal gas exchange.

Table 1. Target-controlled coagulation protocol.

Interval	Parameter	Target	Intervention
Hourly	Haemoglobin	>10 g/dl	Red cell concentrate
Daily	Platelets (citrate)	>100 thousand/ μ l	Platelet concentrate
	INR	<1.35	PPSB
	aPTT	40 - 45 sec	Heparin reduction, Fresh frozen plasma
Monday + Thursday and when clinical bleeding signs are present	Factor VIII	>70 %	10 IU/kg Factor VIII concentrate i.v.
	Factor XIII	>50 %	1250 IU Factor XIII concentrate i.v.
	VWF:Ag	VWF:RCo/VWF:Ag ratio >0.6	0.2 μ g/kg Desmopressin i.v.
	VWF:RCo	VWF:A/VWF:Ag ratio >0.73	→ if target value not reached, repetition of 0.2 μ g/kg Desmopressin i.v.
	since 10/2012: VWF:A		→ if target value still not reached, administration of FVIII+VWF-C (10 IU/kg i.v.)
	RoTEM: Fibrinogen deficiency	FibTEM MCF <10 mm	2 g Fibrinogen i.v.

aPTT: activated partial thromboplastin time; FVIII+VWF-C: Factor VIII + von Willebrand factor concentrate; INR: International normalised ratio; MCF: maximum clot firmness; PPSB: prothrombin complex; RoTEM: rotational thrombelastometry; VWF:Ag: von Willebrand antigen; VWF:RCo: ristocetin cofactor; VWF:A: von Willebrand activity; IU: international units

Protocol

Before the introduction of the coagulation protocol in July 2011, factor concentrates and/or blood products were exclusively given only when signs of bleeding occurred. After the implementation of the coagulation protocol, the administration of clotting factors as well as blood products was performed in a target-controlled approach according to the regimen presented in Table 1. For anticoagulation during ECMO therapy, all patients received a continuous infusion of heparin, with an activated partial thromboplastin time (aPTT) target of 45 - 55 seconds before implementation and a target of 40 - 45 seconds after the implementation of the protocol.

Blood and haemostatic products

To compare the consumption of blood products between the two groups, their amounts were correlated with the bodyweight and the duration of ECMO therapy for each patient (volume/bodyweight/day). The following haemostatic drugs were applied: fibrinogen (factor I) (Haemocomplettan P®, CSL Behring, Marburg, Germany), von Willebrand factor containing factor VIII (Haemate®, CSL Behring), factor XIII (Fibrogammin®, CSL Behring), prothrombin complex (Octaplex®, Octapharma, Langenfeld, Germany), and desmopressin acetate (Minirin®, Ferring Arzneimittel GmbH, Kiel, Germany). Details are presented in Table 1.

Diagnosis of intracranial haemorrhage

All CCT scans from before, during and after the ECMO therapy have been retrospectively evaluated, paying special regard to intracranial haemorrhage.

Statistics

Data were collected in Microsoft Excel sheets (Microsoft, Redmond, Washington, USA). Statistical analysis was performed with GraphPad InStat® (GraphPad Software Inc., La Jolla, California, USA) using Fisher's exact test (incidence of intracranial bleeding) and the Mann-Whitney-U-test (consumption of blood products). A p-value of <0.05 was chosen as the level of significance.

Results

Patient characteristics

Nineteen consecutive patients were included in the control group and 20 consecutive patients in the intervention group. The two groups were comparable in terms of gender, age, bodyweight index and simplified acute physiology score (SAPS) II.¹³ Details are presented in Table 2.

Acquired coagulation disorders during ECMO therapy

As presented in Figure 1, 24 (88%) of all patients measured for factor XIII activity developed factor XIII deficiency, 22 (81%) of them within the first seven days of ECMO therapy. Nineteen (79%) patients measured for AVWS exhibited acquired von Willebrand disease within the first seven days of ECMO. Eight out of 20 patients (40%) in the intervention group exhibited acquired factor I (fibrinogen) generation and stabilisation deficiency in RoTEM analysis. Twenty-one (54%) of all patients in both groups showed a decline of platelet counts greater than 20% in the first 24 hours of ECMO therapy.

Table 2. Patient characteristics.

	Control Group	Intervention Group
Number of patients	19	20
– Male [n (%)]	14 (74)	14 (70)
– Female [n (%)]	5 (26)	6 (30)
Age [y], median (Range)	53 (25 - 66)	53 (23 - 74)
SAPS II, median (Range)	39 (27 - 55)	49 (13 - 79)
BMI [kg/m ²], median (Range)	27.7 (20 - 47)	26.2 (17 - 63)
Cannulation [n (%)]		
– Femoral-jugular	9 (47)	5 (25)
– Femoral-subclavian	1 (5)	–
– Bifemoral	2 (11)	–
– Two-Stage transjugular	7 (37)	14 (70)
– NovaTwin	–	1 (5)
Indication [n (%)]		
– Lung transplant bridging	5 (26)	5 (25)
– Pneumonia, IECOPD	6 (32)	3 (15)
– ARDS following Sepsis	5 (26)	8 (40)
– Aspiration	1 (5)	3 (15)
– Trauma	2 (11)	–
– Severe air leak syndrome	–	1 (5)
Duration of ECMO therapy [d], median (Range)	11 (1 - 51)	9 (2 - 43)
Laboratory results (median of all results of all patients during ECMO therapy)		<i>p-value</i>
– Platelets [thousand/ μ l]	125	119 0.76
– INR	1.08	1.14 0.02
– aPTT [sec]	50.1	40.3 0.002

aPTT: activated partial thromboplastin time; BMI: body mass index, IECOPD: infection exacerbation of chronic obstructive pulmonary disease; INR: international normalised ratio; SAPS II: simplified acute physiology score[9].

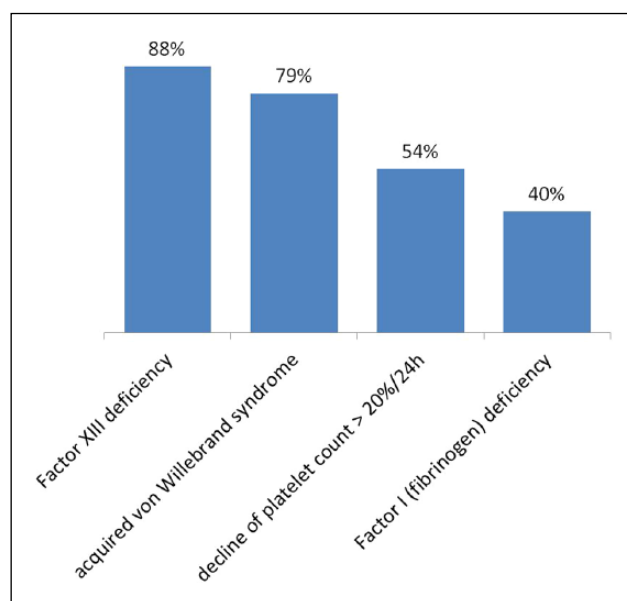
**Figure 1.** Incidence of acquired coagulation disorders in the first 7 days of ECMO therapy.

Table 3. Consumption of blood products during veno-venous ECMO therapy per kgBW.

	Control group	Intervention group
	[ml/kgbw/ECMO days]	
Platelet concentrates		
Mean $\pm$ SEM	1.42 $\pm$ 0.37	3.15 $\pm$ 0.78
Red cell concentrates		
Mean $\pm$ SEM	8.97 $\pm$ 1.76	7.32 $\pm$ 1.60
Fresh frozen plasma		
Mean $\pm$ SEM	1.56 $\pm$ 0.54	3.59 $\pm$ 1.21
	[IU(mg)/kgbw/ECMO days]	
Coagulation factor concentrates		
– Factor VIII + VWF (IU) Mean $\pm$ SEM	0.35 $\pm$ 0.19	0.17 $\pm$ 0.10
– Factor XIII (IU) Mean $\pm$ SEM	1.32 $\pm$ 0.54	1.42 $\pm$ 0.42
– Fibrinogen (mg) Mean $\pm$ SEM	0.59 $\pm$ 0.0	2.81 $\pm$ 1.0
– PPSB (IU) Mean $\pm$ SEM	0 $\pm$ 0	0.07 $\pm$ 0.07

SEM: standard error of mean; kgbw: kilogram bodyweight; VWF: von Willebrand factor; PPSB: prothrombin complex; IU: international unit.

Consumption of blood products and clotting factors

The mean consumption of platelet concentrates, as well as of fresh frozen plasma, factor XIII concentrate and fibrinogen concentrate, was higher in the intervention group. In contrast, the amount of transfused red cell concentrates and von Willebrand factor containing factor VIII was higher in the control group. However, there was no statistically significant difference between the two groups regarding the consumption of blood products and clotting factors. Details are presented in Table 3.

Incidence and severity of intracranial haemorrhage

Six out of 19 (32%) patients in the control group suffered from intracranial haemorrhage. Two of these had severe intracerebral bleeding without previous CCT and became evident by unresponsive pupils. Three of the patients had CCT without bleeding signs during ECMO before intracranial haemorrhage was diagnosed in another CCT scan. One patient had septic embolic cerebral ischaemic areas before, which transformed haemorrhagic during ECMO. After decannulation, this patient recovered. One of the six patients received decompressive hemicraniectomy and died in due course. In four patients, intensive care treatment was discontinued because of fatal neurological prognosis or according to the relatives' decision.

In the intervention group, two patients showed subarachnoid bleeding; one of them had unsuspicious cranial imaging before treatment. Both patients were successfully weaned from ECMO and fully recovered.

Statistical analysis showed a $p=0.13$ using Fisher's exact test to check for differences in the incidence of intracranial bleeding between the groups.

Mortality

In the control group, 12 out of 19 (63%) and in the intervention group 10 out of 20 (50%) patients died. Overall, five patients died due to intracranial bleeding, six because ARDS did not recover or septic shock led to multi-organ failure, five were quoted as "non-transplantable" waiting for lung transplant because of different reasons, one suffered from liver rupture following cardiopulmonary resuscitation (CPR) because of ventricular fibrillation, two had postoperative complications not related to ECMO therapy, one suffered from acute right heart decompensation, in one, a massive bronchial fistula could not be surgically occluded and, in one patient, acute haematological malignancy without therapeutic options was diagnosed. Statistical analysis showed a $p=0.19$, using Fisher's exact test to check for differences in mortality between the groups. No fatal thromboembolic complications, such as high-risk pulmonary embolism during and after ECMO therapy or thrombotic breakdown of the ECMO circuit, have been observed.

Discussion

Veno-venous ECMO is an important option in the treatment of severe respiratory failure and as a bridge-to-lung-transplant procedure. A major drawback of this very invasive therapy, however, is the risk of fatal haemorrhage during extracorporeal therapy. The most threatening and often lethal complication is spontaneous

intracranial bleeding.^{1,2} Understanding the causes and underlying mechanisms leading to coagulation impairment during ECMO therapy would be an important goal to reduce mortality in patients receiving extracorporeal support.

Two mechanisms have been identified so far:

The pump itself, as well as the positive and negative pressures generated in the system, mechanically damage blood cells and, in particular, platelets. This leads to impairment or loss of their function^{14,15} or even lysis of these cells. Contact with the artificial surface of the extracorporeal support system induces a systemic inflammatory response syndrome¹⁶ and diffuse activation of the coagulation system with consecutive wastage of plasmatic coagulation factors. Interestingly, even patients with signs of diffuse clinical bleeding frequently show normal values in standard coagulation tests. This supports our hypothesis that the identification of factors leading to haemorrhage during ECMO therapy by performing standard tests is often unsuccessful and, therefore, bleeding complications are difficult to predict.

In this study, we identified three different pathologies which may contribute to bleeding in ECMO patients. These are destabilisation of the fibrin clot by fibrinogen and factor XIII deficiency, acquired von Willebrand syndrome and thrombocytopenia.

Destabilisation of fibrin clot by fibrinogen and factor XIII deficiency

Clot formation and firmness are dependent on adequate concentrations of factor I (fibrinogen) and factor XIII (fibrin stabilizing factor).¹⁷ Within the present study, we identified factor XIII deficiency in 88% of all patients measured for factor XIII activity. Having in mind that factor XIII deficiency could be responsible for severe bleeding events in the peri-interventional and peri-operative setting and that this pathology cannot be detected by the usual coagulation tests such as aPTT or INR,^{18,19} this is an important new finding. Acquired factor XIII deficiency has also been reported previously in critically ill patients without ECMO therapy. Acquired factor XIII deficiency may also be due to impaired synthesis in the liver (acute liver failure) and increased consumption due to sepsis, major surgery, trauma and disseminated intravascular coagulation.^{20,21} Therefore, we cannot finally prove that factor XIII deficiency is exclusively caused or presumably aggravated by ECMO therapy. However, factor XIII deficiency was present in the majority of our patients and may contribute to an increase in bleeding complications when untreated. The incidence of acquired fibrinogen generation and stabilisation deficiency was 40% in patients tested with the RoTEM/FibTEM method in our study.

Fibrinogen deficiency caused by unspecific activation of the coagulation system and hyperfibrinolysis is a well-known side effect of extracorporeal circulation.

Acquired von Willebrand syndrome

A very important substrate for the activation and aggregation of platelets leading to primary haemostasis is von Willebrand factor (VWF) and, especially, the high molecular weight (HMW) multimers of VWF. Although increased concentrations of VWF antigen are found in patients undergoing veno-venous ECMO therapy, the activity and the amount of VWF HMW multimers constantly decrease, leading to acquired von Willebrand syndrome (AVWS).⁸ We show that 79% of all patients measured for AVWS developed this disorder within the first seven ECMO days. AVWS can be diagnosed only by specific tests that are not routinely performed in patients receiving veno-venous extracorporeal support. Without specific testing, the only sign of an AVWS could often be a moderate prolongation of aPTT, a finding also seen during therapy with heparin. Heparin infusion is currently regarded as the standard anticoagulation method to prevent clotting of the perfusion system. Therefore, aPTT would be prolonged in most cases and intensivists may not even think about the development of AVWS in this situation. The occurrence of AVWS has been described for veno-arterial extracorporeal life support systems (ECLS)¹¹ and ventricular assist devices (VAD)¹⁰ and, recently, also for veno-venous ECMO by our group.⁸ Flow acceleration could be one important reason for shear stress in patients with ECMO devices and lead to AVWS.^{10,11,22,23}

Thrombocytopenia and platelet dysfunction

Fifty-four percent of all patients of both groups showed a rapid decline of platelet counts greater than 20% within the first 24 hours of ECMO therapy. Thrombocytopenia is a common side-effect of critical illness, for various reasons. Shear stress, negative pressure phenomena and diffuse activation of coagulation due to foreign layers additionally lead to increased thrombocytolysis during extracorporeal circulation. Besides decreased platelet counts, the impaired function of residual platelets seems to play a significant role in haemorrhagic diathesis during ECMO therapy.^{14,15,24} Currently, the only effective therapeutic option for platelet dysfunction is the substitution of platelet concentrates, although the resulting effect is transient as well. The optimal target value, however, is still unknown. In this study, a threshold of 100,000/ μ l has been determined, according to national guidelines, for

the treatment of patients with traumatic brain injury. Meanwhile, ELSO also recommends a threshold of 100,000/ μ l, particularly in neonates, in the coagulation guidelines from 2014. Since extracorporeal circuit-induced platelet dysfunction is a major drawback in ECMO therapy, it has to be evaluated if the implementation of new materials or different designs of extracorporeal pumps and oxygenators have the potential to decrease platelet injury in the future.^{25,26}

Heparin

The Extracorporeal Life Support Organisation (ELSO) still recommends to anticoagulate patients during ECMO therapy with heparin, aiming for a PTT of 40-50 sec.²⁷ The optimal target values for balancing anticoagulation therapy or substituting blood products are completely unknown. Amongst others, due to the different designs of commercially available ECMO devices and insufficient statistical power, the prospective randomized trials investigating these issues are still missing. The design and coating of the extracorporeal components seem to play an important role with regard to activation of the coagulation system.^{25,26,28}

Patients in our study received a continuous infusion of heparin, with an aPTT target of 45 - 55 seconds before implementation and a target of 40-45 seconds after implementation of the protocol. Consequently, the median aPTT measured was 50 sec in the control and 40 sec in the intervention group ($p=0.002$). This significant difference in aPTT values between the groups could likely be caused by a more restrictive use of intravenous heparin on the one hand, but also, by a certain degree, could be the result of treatment of von Willebrand syndrome on the other hand. It cannot be excluded that the difference in the incidence of intracranial bleeding events between the groups may be caused, at least in part, by the abovementioned differences in aPTT values (50 vs. 40 sec). However, moderate elevations of aPTT around 50 sec alone are unlikely to be the reason for spontaneous intracranial haemorrhage.

Decrease of intracranial haemorrhage and improvement of neurological outcome

The target-controlled coagulation protocol presented in this manuscript was introduced after the authors had lost four ECMO patients due to fatal intracranial haemorrhage within 14 months. Until that time, the concept of coagulation management was to substitute blood products and factor concentrates only in cases of diffuse bleeding at puncture sites or the mucosa. Implementation and adherence to the new protocol (Table 1) lowered the incidence of intracranial haemorrhage from 31% ($n=6$; before implementation) to 10%

($n=2$; after implementation of the protocol). Intracranial haemorrhage was survived with reasonable neurological outcome in the intervention group while it was massive and led to death in five of six patients in the control group. However, the study population was too small to show statistical significance.

Consumption of blood products

After implementation of the protocol, we found a tendency to a higher amount of platelet concentrates given in the intervention group, without statistical significance. We demonstrated the same trend for FFP, fibrinogen concentrate and PPSB. On the other hand, we observed a higher amount of red cell concentrates given in the control group compared to the intervention group. This effect might be caused by the administration of the abovementioned coagulation products, with the consequent prevention of diffuse bleeding from mucosa or catheter introduction sites and, therefore, diminished blood loss. In any case, the amount of plasma and packed red cells transfused in our study was much lower in both groups compared to the literature.^{9,12}

Conclusions

In this study, we demonstrated that veno-venous ECMO therapy leads to acquired coagulation disorders which cannot be diagnosed using routine coagulation tests. Besides thrombocytopenia, for the first time, we identified factor XIII deficiency as the leading syndrome in 88% of all patients under extracorporeal support. In addition, acquired von Willebrand syndrome was diagnosed in 79% and fibrinogen deficiency in 40% of the patients. The implementation of a coagulation protocol including standardized determination and target-controlled substitution of coagulation factors led to a reduction in the incidence and severity of spontaneous intracranial bleeding from 31% before implementation to 10% after implementation of the protocol. Further research with a larger group of patients is needed to statistically prove this finding. Patients with veno-venous ECMO therapy may profit from screening for factor XIII and fibrinogen deficiency, acquired von Willebrand syndrome and thrombocytopenia. However, the respective thresholds for the administration of factor concentrates, as well as blood products, have to be established in the future.

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Declaration of Conflicting Interest

The authors declare that they have no conflict of interest.

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